preference. In any dynamic model of capital accumulation and growth the concept of golden rule plays an important role. The ***golden rule*** itself refers to that steady state value of capital-labour ratio where $f'(k) = n$, i.e., the marginal product of capital is exactly equal to the rate of growth of population. The significance of the golden rule obtains from the fact that it is that steady state value of capital-labour ratio which corresponds to the maximum steady state value of per capita consumption. Note that when households' rate of time preference is zero, the modified golden rule coincides with the golden rule and the steady state of the Ramsey-Cass-Koopmans model is attained at the golden rule point. However, when people have positive rate of time preference, they are unwilling to sacrifice current consumption for future consumption beyond a point; as a result they accumulate less and reach a lower level of steady state consumption than in the golden rule.

9.3 DECENTRALIZED HOUSEHOLDS' PROBLEM

The Ramsey model described above was transformed by Cass and Koopmans into the problem of a decentralized competitive market economy consisting of many households – all with identical preferences. We shall see that under certain assumptions the optimal solution path in the two cases turns out to be identical.

Instead of assuming the existence of a central planner, let us consider a market economy where the households are price takers. There are two competitive factor markets which determine the wage rate (w_t) and the interest rate or the rate of return on capital (r_t) at each point of time.

There exist many identical competitive firms which hire in labour and capital from the households at the above mentioned wage rate and rental rate. Using labour and capital these firms produce the final output, using the same technology as the central planner. Competition ensures that the wage rate and the rental rate are equal to the respective marginal products of labour and capital at full employment; i.e., $r_t = f'(k_t)$ and $w_t = f(k_t) - k_t f'(k_t)$.

Each household maximizes its welfare given as before. Its objective function is to maximize $W = \int_0^\infty u(c_t)\exp^{-\rho t}\, dt$. Each household starts with a certain amount of capital stock and labour stock. Additionally, households can borrow from one another. Let $a_t \equiv k_t - b_t$ denote the per capita asset stock of an household at period *t*, which consists of the per capita capital stock owned by the household at period *t* minus the amount of debt (per capita) at period *t*. (If the household is a net lender, then b_t would be negative). Arbitrage condition in the asset market ensures that physical capital and lending earns the same rate of return. Hence the per capita income of the household at period *t* is $w_t + r_t a_t$, which the household spends on consumption and further asset accumulation. Thus the budget constraint faced by the household in per capita terms is given by: $c_t + \frac{da}{dt} + na_t = w_t + r_t a_t$.

The complete dynamic optimization problem for the household can now be written in terms of the two time dependent variables: per capita consumption (c_t) and per capita asset stock (a_t). Our optimization problem is as follows:

Maximize $W = \int_0^\infty u(c_t)\exp^{-\rho t}\, dt$

subject to

$$\frac{da}{dt} = w_t + (r_t - n)a_t - c_t$$

In the above problem, a_0 is given exogenously.

The corresponding Hamiltonian function and the first order conditions in terms of the control variable (c_t), state variable (a_t) and co-state variable (λ_t) are as follows:

$$H = u(c_t)\exp^{-\rho t} + \lambda_t\left[w_t + (r_t - n)a_t - c_t\right]$$

(ib) $u'(c_t)\exp^{-\rho t} = \lambda_t$

(iib) $\frac{d\lambda}{dt} = -\lambda_t(r_t - n)$

(iiib) $\frac{da}{dt} = w_t + (r_t - n)a_t - c_t$

(ivb) $\lim_{t\to\infty} \lambda_t a_t = 0$

Note that these conditions look quite similar to the first order conditions that we obtained for the central planner's problem (conditions (ia)-(iva) in the previous section). However, for the household's problem there is an additional condition which has to be satisfied if we want to get a meaningful solution. This condition is called the ***No-Ponzi-Game condition*** which we elaborate below. First note that we have allowed for intra-household borrowing which means that a household can maintain a consumption stream above its income by borrowing. Now if a household could go on borrowing indefinitely then it will be optimal for the household to always maintain an infinitely high (or maximum possible) consumption stream and finance such high level of consumption simply by borrowing more and more. Of course, such a strategy would also imply that the net per capita borrowing of the household would increase exponentially at the rate $(r_t - n)$ (since the household has to borrow not only for consumption, but also to pay back the interest rate as well) and the present discounted value of the net debt of the household will approach infinity. Such a financing scheme is called Ponzi-Game financing. In order to rule out such infinite indebtedness by any family we specify the condition that the even though the household can be temporarily a net debtor (i.e., the present discounted value of its asset is temporarily negative), over a sufficiently longer horizon, it must eventually repay all its debt and hold non-negative asset stock. Formally, as *t* goes to infinity, the present discounted value of the (per capita) asset stock of the household must be non-negative:

$$\text{(vb)}\ \lim_{t\to 0} a_t \exp^{-\int_0^t (r_\tau - n)d\tau} \geq 0\,.$$

Condition (vb) is called the No-Ponzi-Game condition. For a decentralized household conditions (ib)-(vb) specify the optimal path that solves the household's dynamic optimization problem.

As before, differentiating (ib) with respect to *t* and using (iib) to eliminate λ_t, and noting that the term $\frac{-cu''(c)}{u'(c)} \equiv \sigma$ denotes the elasticity of marginal utility with respect to consumption, we can derive the following differential equation:

$$\text{(vib)}\ \frac{dc}{dt} = \left(\frac{c_t}{\sigma}\right)\left(r_t - n - \rho\right)$$

Equations (iiib) and (vib) together form a system of differential equations which, along with the transversality condition and the no-Ponzi-Game condition, determine the movements of per capita consumption and per capita asset stock of the household along the optimal path.

Note that while for the economy as a whole the wage rate and the rental rate are given by $r_t = f'(k_t)$ and $w_t = f(k_t) - k_t f'(k_t)$, the price-taking households take these as given while optimizing. However, we assume that the households are endowed with perfect foresight. It implies that households ***do not know*** the relationship between *w* and *r* with the per capita capital stock; but they can exactly ***guess*** the values of *w* and *r* at every point of time.

Also note that while the households can borrow from one another, in equilibrium for the economy as a whole, net borrowing must be zero. Since households are all identical, this implies that net borrowing of each household would be zero, i.e., $a_t = k_t$.

Thus, replacing $r_t = f'(k_t)$, $w_t = f(k_t) - k_t f'(k_t)$, and $a_t = k_t$ in equations (iiib) and (vib), we get a system of differential equations which is exactly identical to the system of differential equations that we obtained for the central planner. Hence, their solution must also be the same. In other words, under the assumption of perfect foresight on the part of the households, *the household's optimal consumption and accumulation paths are exactly identical to the consumption and accumulation paths that would be chosen by a social planner*. To put it differently, the decentralized market economy's solution path coincides with the solution path of the centralized command economy, and the solution paths can once again characterized by the diagram given earlier. This is a very strong result which shows the equivalence between the market equilibrium and the socially optimal solution. This equivalence result, however, depends crucially on the assumption of perfect foresight.

Check Your Progress 1

1. What are the steady state values of per capita consumption and per capita capital stock in the centralized version of the Ramsey optimal growth model?

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2. Would the above values change if we consider a decentralized economy instead of a centralized planner's problem? Why?

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9.4 GOVERNMENT IN THE RAMSEY-CASS-KOOPMANS MODEL: RICARDIAN EQUIVALENCE

The decentralized equilibrium that we discussed above does not incorporate government as a separate economic agent. It is however easy to introduce a role for the government in the de-centralized Ramsey-Cass-Koopmans model. There are two options to finance government spending: (i) taxation, and (ii) borrowing. Suppose the government finances its expenditure by imposing a lump-sum tax on the household's income in every period. Since the government expenditure is unrelated to the households' consumption expenditure, the utility function of the households remains unchanged. However, since the government is taking away a part of the household's income, the budget constraint of the household changes. To be more precise, if the government imposes a tax τ on the household income then the disposable income of the household reduces to $w_t + r_t a_t - \tau$. Consequently, the budget constraint of the household becomes: $\frac{da}{dt} = w_t + (r_t - n)a_t - c_t - \tau$. The rest of the optimal conditions, discussed in Section 9.3, remains the same.

In the phase diagram therefore, the $\frac{dk}{dt}=0$ curve shifts downward by the amount τ, as shown in Fig. 9.4. The new steady state is obtained at the point E' in Fig. 9.4, where the steady state value of the per capita capital stock remains the same, but the steady state value of per capita consumption is lower. The corresponding optimal path of the household is once again denoted by the dotted line in the diagram.

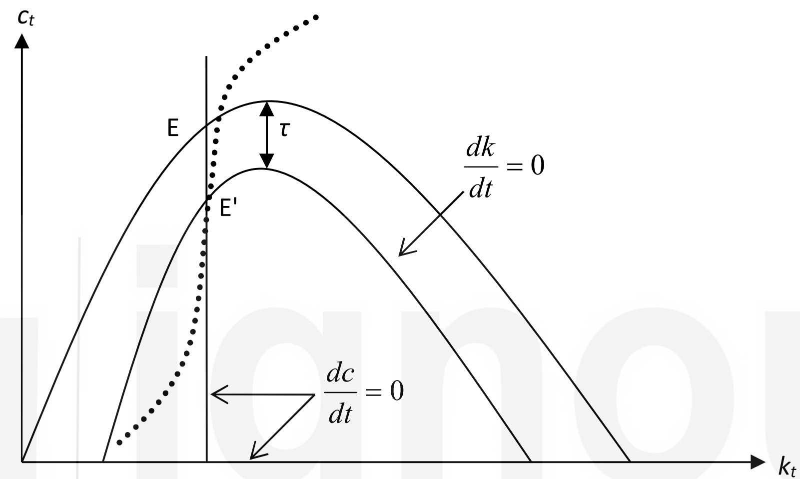

Fig. 9.4: Phase Diagram (with Government)

Now suppose the government decides to finance its expenditure by borrowing from the households, instead of taxation. Will that make any difference to the households' optimal condition? At first glance it seems that if the government borrows from the households by distributing government bonds, then the government bonds would constitute an asset for the households (because the government would pay an interest on these loans). So the household's disposable income is not reduced; only the composition of its asset portfolio changes. So at first glance it seems that the household's optima under tax financing and under debt financing would be different. However, a more careful consideration of the problem reveals that the mode of financing of the government expenditure really **does not** make any difference to the household's optima. As long as the government is forced to maintain a balanced budget in the long run (i.e., it cannot go on borrowing indefinitely), at some point of time in future the government has to impose a tax in order to repay its debt incurred earlier. Since the households have perfect foresight, they would know that at some point of time they would have to pay the tax. Thus, even though households are not paying the taxes today, they will adjust their current consumption and capital accumulation decisions accordingly which will enable them to bear the tax burden in future. This is a strong result: It states that for households endowed with perfect foresight (or rational expectation), the mode of financing of government expenditure does not matter. Tax financing and debt financing are exactly equivalent as far as the household optima is concerned. This result is known as the ***Ricardian Equivalence***.

Check Your Progress 2

1. In the decentralized version of the optimal growth model, how would the equilibrium values of per capita capital stock and the per capita consumption change if the government imposes a lump sum tax to finance its own expenditure? In this context, explain the Ricardian Equivalence proposition.

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9.5 LET US SUM UP

In this unit, we have looked at the Ramsey-Cass-Koopmans optimal growth model where infinitely-lived individuals maximize their life-time utility. We identify the optimal consumption-accumulation path as the saddle path which takes the economy to its steady state in the long run. The steady state point is identified with the modified golden rule point. Although the problem is identified as a social planner's problem, its basic conclusion remains unchanged even when the economy is characterized by competitive decentralized households. Moreover introducing a government sector which finances its expenditure through lump sum taxation does not have any effect on the steady state values of the model.

9.6 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercises 1

1. See Fig. 9.3. The steady state values of per capita consumption and per capita capital stock is given by the intersection of the equations $f'(k) = n + \rho$ and $c = f(k) - nk$.

2. The steady state values of per capita consumption and per capita capital stock are the same in both central planner and decentralized household problems

Check Your Progress Exercises 2

1. If the government imposes a lump sum tax, the steady state value of per capita capital stock remains unchanged while the per capita consumption would be lower. See Section 9.4 for details.

9.7 MATHEMATICAL APPENDIX

Dynamic Optimization in Continuous Time (Optimal Control)

A.1 Finite Horizon Problem

Consider the following optimization problem which is defined over a finite time horizon from 0 to T:

$$\text{Max. } W = \int_0^T F(u_t, x_t, t)\, dt, \text{ subject to } \frac{dx}{dt} = f(u_t, x_t, t) \qquad \ldots(1)$$

$$x_0 = \bar{x}; \; u_t \in U.$$

The objective function here is an ***integral***, and our task is to find out a time path of the time dependent variable u from the choice set U, (i.e., to choose a $u \in U$ for each point of time t starting from 0 to T) such that this integral is maximized.

But our choice is not unconstrained. (Had it been so, a simple point-by-point static optimization exercise would have given us the required solution path). Note that the F function (which can be called the ***instantaneous*** objective function) depends not only on u but also on another time dependent variable x. And our choice of u at each point of time affects the next period's value of x through the given differential equation. Thus our choice of u affects the instantaneous objective function directly, as well as indirectly through x. So when we are choosing the optimal time path for u that maximizes the integral W, we have to take into account how x is changing due to our choice of u – this explains the presence of a constraint function in the form of a differential equation in x.

Since we are choosing the variable u directly (i.e., we have direct control over this variable) it is called the ***control variable***. The other time dependant variable, x, which changes over time due to our choice of u, but we don't have direct control over its change, is called the ***state variable***. The function f in the given differential equation that describes the change in x over time is called the ***state transition function***.

Pontryagin's Maximum Principle: Let u_t^* be a solution path to the problem specified in (1), and let x_t^* be the associated path for the state variable, where u_t^* is a piece-wise continuous function of t and x_t^* is a strictly continuous but piece-wise differentiable in t. Then there exist a strictly continuous and piece-wise differentiable variable λ_t, and a function H defined as:

$$H(u, x, \lambda, t) = F(u_t, x_t, t) + \lambda_t f(u_t, x_t, t),$$

such that

(i) H is maximized with respect to u at u^* for all $t \in [0, T]$

(ii) $$\left.\frac{\partial H}{\partial x}\right|_{(u^*, x^*, \lambda, t)} = -\frac{d\lambda}{dt}$$

(iii) $\left.\dfrac{\partial H}{\partial \lambda}\right|_{(u^*,x^*,\lambda,t)} = \dfrac{dx}{dt}$

(iv) $\lambda_T x_T = 0$

The function H is called the ***Hamiltonian Function*** associated with the given dynamic optimizations problem, and the newly introduced time dependent variable λ is called the ***co-state variable*** associated with the state variable x.

The co-state variable λ_t measures the change in the value of the objective function W associated with an infinitesimal change in the state variable x at time t. If there were an exogenous tiny increment to the state variable at time t, and if the problem were modified optimally there after, then the increment in the total value of the objective would be λ_t. Thus it is the marginal valuation of the state variable at time t, and is therefore often referred to as the ***shadow price*** of the state variable at time t.

Pontryagin's Maximum Principle gives us four ***first order necessary conditions*** for the optimization problem defined in (1). The first three first order necessary conditions (F.O.N.C) are defined in terms of the Hamiltonian function. Note that if the Hamiltonian function is non-linear in u, then (i) can be replaced by the condition $\left.\dfrac{\partial H}{\partial u}\right|_{(u^*,x,\lambda,t)} = 0$, **provided the second order check** $\left.\dfrac{\partial^2 H}{\partial u^2}\right|_{(u^*,x,\lambda,t)} < 0$ **is verified**.

The last condition of the Maximum Principle (condition (iv)), which specifies a terminal condition for λ, is called the ***Transversality Condition***.

A.2 Infinite Horizon Problems

Consider the following infinite horizon dynamic optimization exercise:

$$\text{Max. } W = \int_0^\infty F(u_t, x_t, t)\,dt, \text{ subject to } \frac{dx}{dt} = f(u_t, x_t, t) \qquad \ldots(2)$$

$$x_0 = \bar{x};\ \ u_t \in U.$$

This problem is almost the same as the problem specified in (1); only now the time horizon is no longer finite.

When the time horizon stretches from 0 to ∞, additional complications may arise: First, the integral W may not converge. In that case it does not make sense to talk about optimal paths, because any paths (u_t, x_t) will eventually make the integral infinitely large, and no comparison between one set of (u_t, x_t) with another is possible. Hence, for the dynamic optimization exercise to be meaningful, we must impose additional restrictions so that the integral converges to a finite value.

Second, there is some controversy regarding the appropriate transversality condition in infinite horizon problems. Usually some limiting conditions, which are similar to the transversality conditions applied in the finite horizon problems, are applied.

Consider problem (2) and suppose the integral converges. The F.O.N.C.s for this problem are given in the following theorem:

Let u_t^* be a solution path to the problem specified in (5), and let x_t^* be the associated path for the state variable, where u_t^* is a piece-wise continuous function of t and x_t^* is a strictly continuous but piece-wise differentiable in t. Then there exist a strictly continuous and piece-wise differentiable variable λ_t, and a function H defined as:

$$H\left(u,x,\lambda,t\right) = F(u_t,x_t,t) + \lambda_t f(u_t,x_t,t),$$

such that

(i) H is maximized with respect to u at u^* for all $t \in [0,\infty)$

(ii) $\left.\dfrac{\partial H}{\partial x}\right|_{(u^*,x^*,\lambda,t)} = -\dfrac{d\lambda}{dt}$

(iii) $\left.\dfrac{\partial H}{\partial \lambda}\right|_{(u^*,x^*,\lambda,t)} = \dfrac{dx}{dt}$

(iv) $\lim\limits_{t\to\infty} \lambda_t x_t = 0$

UNIT 10 OVERLAPPING GENERATIONS MODEL*

Structure

10.0 OBJECTIVES

After going through this Unit you should be able to exlain

- the standard two-period overlapping generations model with production;
- the concept of dynamic efficiency and examine whether the dynamic efficiency property holds for an overlapping generations economy; and
- the role of social security system in the overlapping generations framework in eliminating dynamic inefficiency.

10.1 INTRODUCTION

In the context of intertemporal decision making on the part of the households, you have come across the optimal growth model in Unit 9. There is another important framework that also considers households' intertemporal decision making – although over finite time horizon. This framework is called the overlapping generations framework. The framework was first developed by Samuelson (1954) which has subsequently been widely used in macro dynamics.

In the overlapping generations framework, the individuals have a finite time horizon (in the standard case, a two-period time horizon), but the society lives forever. The term 'overlapping generations' implies that at each point of time the life-time of two generations overlaps. To clarify, let us take the standard case where each generation lives exactly for two periods. Thus, for the cohort of individuals who are born at the beginning of period 't', they are alive in period 't' (when they are young) and in period $(t+1)$ (when they are old). On the other hand, a new set of individuals are born at the beginning of period $(t+1)$, who would be alive in period $(t+1)$ and period $(t+2)$. Thus, between the lifetimes of these two successive generations, there is an exact overlap of one period. In the following discussion we shall concentrate on the two-period overlapping generations framework only.

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